

Script started on 2026-02-17 18:23:47-06:00 [TERM="xterm" TTY="/dev/pts/36" COLUMNS=100]
rc47379@ares:~\$ pwd

/home/students/rc47379
rc47379@ares:~\$ cat rolodex.info

NAME: Calvin Rice

CLASS: CSC122-002

TITLE: Rolodex

LEVEL: 4

DESCRIPTION:

A list of people's names associated with their address and phone number that has s

Unknown class/file type! Please have your teacher request a group update...
rc47379@ares:~\$ show-code rolodex.cpp

rolodex.cpp:

```
1 #include <iostream>
2 #include <ostream>
3 #include <vector>
4 #include <cctype>
5 #include <algorithm>
6
7 #include "inputprotection.h"
8 using namespace std;
9
10 class RoloEntry
11 {
12     private:
13         string name, street, city, state, phone;
14
15     public:
16         RoloEntry()
17         {
18             name = "EMPTY";
19             street = "EMPTY";
20             city = "EMPTY";
21             state = "EMPTY";
22             phone = "EMPTY";
23         }
24
25         RoloEntry(string _name, string _street, string _city, string _state, s
26         {
27             name = _name;
28             street = _street;
```

```
29         city = _city;
30         state = _state;
31         phone = _phone;
32     }
33
34     string get_name()
35     {
36         return name;
37     }
38
39     string get_street()
40     {
41         return street;
42     }
43
44     string get_city()
45     {
46         return city;
47     }
48
49     string get_state()
50     {
51         return state;
52     }
53
54     string get_phone()
55     {
56         return phone;
57     }
58
59     void set_name(string newName)
60     {
61         name = newName;
62     }
63
64     void set_street(string newStreet)
65     {
66         street = newStreet;
67     }
68
69     void set_city(string newCity)
70     {
71         city = newCity;
72     }
73
74     void set_state(string newState)
75     {
76         state = newState;
77     }
78
79     void set_phone(string newPhone)
80     {
81         phone = newPhone;
82     }
```



```

191         string newState;
192         getline(cin, newState);
193         selectedEntry.set_state(newState);
194         break;
195     }
196     case 5:
197     {
198         cout << "Current Phone Number: " << selectedEntry.get_phone();
199         cout << "Enter a New Phone Number: ";
200         string newPhone;
201         getline(cin, newPhone);
202         selectedEntry.set_phone(newPhone);
203         break;
204     }
205     case 6:
206     {
207         break;
208     }
209     default:
210     {
211         break;
212     }
213     rolodex[entrySelectNumber - 1] = selectedEntry;
214     break;
215 case '3':
216 {
217     for(int i = 0; i < rolodex.size(); i++)
218     {
219         cout << i + 1 << " ) " << rolodex.at(i).get_name() << " ";
220     }
221     cout << endl;
222     long entrySelectNumber = input_protect("Select which entry to edit: ");
223     if(entrySelectNumber == 0)
224     {
225         break;
226     }
227     entrySelectNumber -= 1;
228     rolodex.erase(rolodex.begin() + entrySelectNumber);
229     break;
230 }
231 case '4':
232 {
233     cout << "1) Find by Name"
234           << "\n2) Find by Address"
235           << "\n3) Find by Phone number"
236           << "\n4) Cancel" << endl;
237     char findCategory = input_protect("Select a category: ");
238     if(findCategory == '1')
239     {
240         cout << "Enter query to search for: " << endl;
241         string query;
242         getline(cin, query);
243         string queryLower = query;

```

```

245         for(char& c : queryLower)
246         {
247             c = tolower(c);
248         }
249         vector<RoloEntry> results = {};
250         switch(findCategory)
251         {
252             case '1':
253             {
254                 for(int i = 0; i < rolodex.size(); i++)
255                 {
256                     string nameLower = rolodex.at(i).get_name();
257                     for(char& c : nameLower)
258                     {
259                         c = tolower(c);
260                     }
261                     if(nameLower.find(query) != std::string::npos)
262                     {
263                         results.push_back(rolodex.at(i));
264                     }
265                 }
266                 break;
267             }
268             case '2':
269             {
270                 for(int i = 0; i < rolodex.size(); i++)
271                 {
272                     string addressLower = rolodex.at(i).get_address();
273                     for(char& c : addressLower)
274                     {
275                         c = tolower(c);
276                     }
277                     if(addressLower.find(query) != std::string::npos)
278                     {
279                         results.push_back(rolodex.at(i));
280                     }
281                 }
282                 break;
283             }
284             case '3':
285             {
286                 for(int i = 0; i < rolodex.size(); i++)
287                 {
288                     if(rolodex.at(i).get_phone().find(query) != std::string::npos)
289                     {
290                         results.push_back(rolodex.at(i));
291                     }
292                 }
293                 break;
294             }
295             case '4':
296             {
297                 break;
298             }
299             default:
300             {
301                 break;
302             }
303         }
304         if(findCategory != '4')

```

```

299         {
300             cout << "\nFound " << results.size() << " results r
301             for(int i = 0; i < results.size(); i++)
302             {
303                 cout << i + 1 << " ) " << results.at(i).get_name()
304                 cout << results.at(i).get_street() << endl;
305                 cout << results.at(i).get_city() << ", " << re
306                 cout << results.at(i).get_phone() << endl << ei
307             }
308         }
309         break;
310     }
311     case '5':
312     {
313         cout << "All Entries: " << endl << endl;
314         for(int i = 0; i < rolodex.size(); i++)
315         {
316             cout << i + 1 << " ) " << rolodex.at(i).get_name() <
317             cout << rolodex.at(i).get_street() << endl;
318             cout << rolodex.at(i).get_city() << ", " << rolode
319             cout << rolodex.at(i).get_phone() << endl << endl;
320         }
321         break;
322     }
323     case '6':
324         return 0;
325     default:
326         break;
327 }
328 }
329
330 return 0;
331 }

```

rc47379@ares:~\$ show-code inputprotection.cpp

inputprotection.cpp:

```

1  #include <iostream>
2  #include <string>
3  #include "inputprotection.h"
4
5  double input_protect(std::string prompt, std::string errorMessage, double l
6  {
7      while(true)
8      {
9          std::cout << prompt << std::endl;
10         double input;
11         std::cin >> input;
12         std::cin.ignore(std::numeric_limits<std::streamsize>::max(), '\n');
13
14         if(input < lowerBound || input > upperBound)

```

```

15         {
16             std::cout << errorMessage << std::endl;
17         }
18         else
19         {
20             return input;
21         }
22     }
23 }
24 double input_protect(std::string prompt, std::string errorMessage, double l
25 {
26     while(true)
27     {
28         std::cout << prompt << std::endl;
29         double input;
30         std::cin >> input;
31         std::cin.ignore(std::numeric_limits<std::streamsize>::max(), '\n');
32
33         if(input > bound && upper)
34         {
35             std::cout << errorMessage << std::endl;
36         }
37         else if(input < bound && !upper)
38         {
39             std::cout << errorMessage << std::endl;
40         }
41         else
42         {
43             return input;
44         }
45     }
46 }
47 }
48
49 long input_protect(std::string prompt, std::string errorMessage, long upper
50 {
51     while(true)
52     {
53         std::cout << prompt << std::endl;
54         long input;
55         std::cin >> input;
56         std::cin.ignore(std::numeric_limits<std::streamsize>::max(), '\n');
57
58         if(input < lowerBound || input > upperBound)
59         {
60             std::cout << errorMessage << std::endl;
61         }
62         else
63         {
64             return input;
65         }
66     }
67 }
68 }

```

```

69 long input_protect(std::string prompt, std::string errorMessage, long bound
70 {
71     while(true)
72     {
73         std::cout << prompt << std::endl;
74         long input;
75         std::cin >> input;
76         std::cin.ignore(std::numeric_limits<std::streamsize>::max(), '\n');
77
78         if(input > bound && upper)
79         {
80             std::cout << errorMessage << std::endl;
81         }
82         else if(input < bound && !upper)
83         {
84             std::cout << errorMessage << std::endl;
85         }
86         else
87         {
88             return input;
89         }
90     }
91 }
92
93 char input_protect(std::string prompt, std::string errorMessage, std::string
94 {
95     bool invalid = true;
96     while(true)
97     {
98         std::cout << prompt << std::endl;
99         char input;
100         std::cin >> input;
101         std::cin.ignore(std::numeric_limits<std::streamsize>::max(), '\n');
102
103         for(char c : validChoices)
104         {
105             if(c == input)
106             {
107                 return input;
108             }
109         }
110
111         std::cout << errorMessage << std::endl;
112     }
113 }
114
115 }

```

rc47379@ares:~\$ show-code inputprotection.h

inputprotection.h:

```

1 #ifndef INPUT_PROT
2 #define INPUT_PROT
3 #include <string>
4
5 double input_protect(std::string prompt, std::string errorMessage, double l
6 double input_protect(std::string prompt, std::string errorMessage, double l
7
8 long input_protect(std::string prompt, std::string errorMessage, long upper
9 long input_protect(std::string prompt, std::string errorMessage, long bound
10
11 char input_protect(std::string prompt, std::string errorMessage, std::stri
12 #endif

```

rc47379@ares:~\$ CPP rolodex inputprotection

inputprotection.cpp...

rolodex.cpp***

inputprotection.cpp: In function

'char input_protect(std::string, std::string, std::string)':

inputprotection.cpp:96:10: warning: unused variable 'invalid' [-Wunused-variable]

```

96 |     bool invalid = true;
    |         ^~~~~~

```

rolodex.cpp: In constructor

'RoloEntry::RoloEntry()':

rolodex.cpp:17:5: warning:

'RoloEntry::name' should be initialized in the member initialization list [-Weffc++]

```

17 |     RoloEntry()
    |     ^~~~~~

```

rolodex.cpp:17:5: warning:

'RoloEntry::street' should be initialized in the member initialization list [-Weffc++]

rolodex.cpp:17:5: warning:

'RoloEntry::city' should be initialized in the member initialization list [-Weffc++]

rolodex.cpp:17:5: warning:

'RoloEntry::state' should be initialized in the member initialization list [-Weffc++]

rolodex.cpp:17:5: warning:

'RoloEntry::phone' should be initialized in the member initialization list [-Weffc++]

rolodex.cpp: In constructor

'RoloEntry::RoloEntry(std::string, std::string, std::string, std::string)':

rolodex.cpp:26:5: warning:

'RoloEntry::name' should be initialized in the member initialization list [-Weffc++]

```

26 |     RoloEntry(string_name, string_street,
    |             ^~~~~~
    |             string_city, string_state, string_phone)

```

rolodex.cpp:26:5: warning:

'RoloEntry::street' should be initialized in

```
the member initialization list [-Weffc++]
rolodex.cpp:26:5: warning:
'RoloEntry::city' should be initialized in
the member initialization list [-Weffc++]
rolodex.cpp:26:5: warning:
'RoloEntry::state' should be initialized in
the member initialization list [-Weffc++]
rolodex.cpp:26:5: warning:
'RoloEntry::phone' should be initialized in
the member initialization list [-Weffc++]
rolodex.cpp: In function 'int
main()':
rolodex.cpp:132:38: warning: comparison
of integer expressions of different signedness: 'int'
and 'std::vector<RoloEntry>::size_type'
{aka 'long unsigned int'}
[-Wsign-compare]
132 |           for(int i = 0; i <
      |           rolodex.size(); i++)
      |           ~~~~~^~~~~~
rolodex.cpp:218:38: warning: comparison
of integer expressions of different signedness: 'int'
and 'std::vector<RoloEntry>::size_type'
{aka 'long unsigned int'}
[-Wsign-compare]
218 |           for(int i = 0; i <
      |           rolodex.size(); i++)
      |           ~~~~~^~~~~~
rolodex.cpp:248:36: warning: conversion
from 'int' to 'char' may change
value [-Wconversion]
248 |           c = tolower(c);
      |           ~~~~~^~~~~~
rolodex.cpp:255:46: warning: comparison
of integer expressions of different signedness: 'int'
and 'std::vector<RoloEntry>::size_type'
{aka 'long unsigned int'}
[-Wsign-compare]
255 |           for(int i = 0; i <
      |           rolodex.size(); i++)
      |           ~~~~~^~~~~~
rolodex.cpp:260:48: warning: conversion
from 'int' to 'char' may change
value [-Wconversion]
260 |           c =
      |           ~~~~~^~~~~~
rolodex.cpp:270:46: warning: comparison
of integer expressions of different signedness: 'int'
and 'std::vector<RoloEntry>::size_type'
```

```
{aka 'long unsigned int'}
[-Wsign-compare]
270 |           for(int i = 0; i <
      |           rolodex.size(); i++)
      |           ~~~~~^~~~~~
rolodex.cpp:275:48: warning: conversion
from 'int' to 'char' may change
value [-Wconversion]
275 |           c =
      |           ~~~~~^~~~~~
rolodex.cpp:285:46: warning: comparison
of integer expressions of different signedness: 'int'
and 'std::vector<RoloEntry>::size_type'
{aka 'long unsigned int'}
[-Wsign-compare]
285 |           for(int i = 0; i <
      |           rolodex.size(); i++)
      |           ~~~~~^~~~~~
rolodex.cpp:302:42: warning: comparison
of integer expressions of different signedness: 'int'
and 'std::vector<RoloEntry>::size_type'
{aka 'long unsigned int'}
[-Wsign-compare]
302 |           for(int i = 0; i <
      |           results.size(); i++)
      |           ~~~~~^~~~~~
rolodex.cpp:315:38: warning: comparison
of integer expressions of different signedness: 'int'
and 'std::vector<RoloEntry>::size_type'
{aka 'long unsigned int'}
[-Wsign-compare]
315 |           for(int i = 0; i <
      |           rolodex.size(); i++)
      |           ~~~~~^~~~~~

rc47379@ares:~$ ./rolodex.out

1) Add entry
2) Edit entry
3) Delete entry
4) Find entry
5) Print all entries
6) Quit
Select an Option:
5

All Entries:
```

1) Alice Oneson
27324 Walfred Way
Springfield, IL
217-152-1234

2) Brian Twosome
1095 University Rd
Madison, WI
608-893-5678

3) Gordon Threeman
789 Pine Rd
Austin, TX
512-693-9012

4) David Fourman
749 Harrison Ave
Seattle, WA
206-112-3456

5) Emma Fiveson
4100 University Ave
Denver, CO
303-064-7890

6) Frank Sixman
955 County Rd
Chicago, IL
407-378-2345

1) Add entry
2) Edit entry
3) Delete entry
4) Find entry
5) Print all entries
6) Quit
Select an Option:
1

Enter name: Calvin Rice
Enter street: 127 Street St
Enter city: Arlington Heights
Enter state: IL
Enter phone number: 777-777-7777

1) Add entry
2) Edit entry
3) Delete entry
4) Find entry
5) Print all entries
6) Quit
Select an Option:
5

All Entries:

1) Alice Oneson
27324 Walfred Way
Springfield, IL
217-152-1234

2) Brian Twosome
1095 University Rd
Madison, WI
608-893-5678

3) Gordon Threeman
789 Pine Rd
Austin, TX
512-693-9012

4) David Fourman
749 Harrison Ave
Seattle, WA
206-112-3456

5) Emma Fiveson
4100 University Ave
Denver, CO
303-064-7890

6) Frank Sixman
955 County Rd
Chicago, IL
407-378-2345

7) Calvin Rice
127 Street St
Arlington Heights, IL
777-777-7777

1) Add entry
2) Edit entry
3) Delete entry
4) Find entry
5) Print all entries
6) Quit
Select an Option:
2

1) Alice Oneson
2) Brian Twosome
3) Gordon Threeman
4) David Fourman
5) Emma Fiveson
6) Frank Sixman
7) Calvin Rice
Select which entry you would like to edit, or enter 0 to return to the menu.

5
What would you like to edit?
1) Name
2) Street
3) City
4) State
5) Phone Number
6) CancelEnter a number 1-6:
1
Current Name: Emma Fiveson
Enter a New Name: John Editson
1) Add entry
2) Edit entry
3) Delete entry
4) Find entry
5) Print all entries
6) Quit
Select an Option:
5

All Entries:

- 1) Alice Oneson
27324 Walfred Way
Springfield, IL
217-152-1234
- 2) Brian Twosome
1095 University Rd
Madison, WI
608-893-5678
- 3) Gordon Threeman
789 Pine Rd
Austin, TX
512-693-9012
- 4) David Fourman
749 Harrison Ave
Seattle, WA
206-112-3456
- 5) John Editson
4100 University Ave
Denver, CO
303-064-7890
- 6) Frank Sixman
955 County Rd
Chicago, IL
407-378-2345
- 7) Calvin Rice
127 Street St

Arlington Heights, IL
777-777-7777

- 1) Add entry
 - 2) Edit entry
 - 3) Delete entry
 - 4) Find entry
 - 5) Print all entries
 - 6) Quit
- Select an Option:
4

- 1) Find by Name
 - 2) Find by Address
 - 3) Find by Phone number
 - 4) Cancel
- Select a category:
1
Enter query to search for:
man

Found 3 results matching "man":
1) Gordon Threeman
789 Pine Rd
Austin, TX
512-693-9012

2) David Fourman
749 Harrison Ave
Seattle, WA
206-112-3456

3) Frank Sixman
955 County Rd
Chicago, IL
407-378-2345

- 1) Add entry
 - 2) Edit entry
 - 3) Delete entry
 - 4) Find entry
 - 5) Print all entries
 - 6) Quit
- Select an Option:
3

- 1) Alice Oneson
- 2) Brian Twosome
- 3) Gordon Threeman
- 4) David Fourman
- 5) John Editson
- 6) Frank Sixman
- 7) Calvin Rice

Select which entry you would like to delete, or enter 0 to return to the menu.


```
4
1) Add entry
2) Edit entry
3) Delete entry
4) Find entry
5) Print all entries
6) Quit
Select an Option:
5

All Entries:

1) Alice Oneson
27324 Walfred Way
Springfield, IL
217-152-1234

2) Brian Twosome
1095 University Rd
Madison, WI
608-893-5678

3) Gordon Threeman
789 Pine Rd
Austin, TX
512-693-9012

4) John Editson
4100 University Ave
Denver, CO
303-064-7890

5) Frank Sixman
955 County Rd
Chicago, IL
407-378-2345

6) Calvin Rice
127 Street St
Arlington Heights, IL
777-777-7777

1) Add entry
2) Edit entry
3) Delete entry
4) Find entry
5) Print all entries
6) Quit
Select an Option:
6

rc47379@ares:~$ exit

exit
```

Script done on 2026-02-17 18:26:01-06:00 [COMMAND_EXIT_CODE="0"]